

CAPSTONE PROJECT — AI/ML + CYBERSECURITY

1. AI-Based Multi-Stage Social Engineering Detector

Idea:

An AI system that analyzes complete conversations/emails and detects social-engineering attacks.

What makes it different:

Instead of checking only whether one message is phishing, it identifies the **manipulation tactics and progression** of an attack.

Example:

Trust building → Impersonation → Urgency/Fear → Information request → OTP/Payment request

AI/ML: NLP, BERT/Transformers, intent detection, classification.

Novelty Possibility:

Conversation-level detection + attack-stage detection + explainable risk score + English/Hinglish support.

Difficulty: Medium–High

Potential: Strong real-world + AI + cybersecurity project.

2. AI Prompt Injection Firewall

Idea:

A security layer that protects LLMs/AI agents from malicious instructions hidden in prompts, websites, emails or documents.

What makes it different:

The system doesn't only check the prompt; it also checks the **context and actions the AI agent is trying to perform**.

Example:

Malicious instruction hidden inside a document → AI tries to access/send sensitive data → Firewall detects risk → Action blocked.

AI/ML: NLP, embeddings, ML classification, LLMs, anomaly detection.

Novelty Possibility:

Context-aware detection + multi-stage attack detection + tool/action-level security + hybrid ML/rule-based defense.

Difficulty: High

Potential: Very high research and industry relevance.

3. Multimodal AI Deepfake & Identity Attack Detector

Idea:

An AI system that detects whether someone is being impersonated using fake video, voice, face or other AI-generated media.

What makes it different:

Instead of detecting only a fake video, it combines **face + voice + video + context** to determine identity-attack risk.

Example:

Fake CEO voice + fake video + urgent payment request → System detects possible identity impersonation.

AI/ML: Computer Vision, CNN/Transformers, audio analysis, deepfake detection, multimodal AI.

Novelty Possibility:

Multimodal verification + identity consistency + social-engineering analysis.

Difficulty: Very High

Potential: Extremely impressive but technically challenging.

4. AI-Based Insider Threat & Account Compromise Detection

Idea:

An AI system that learns normal user behavior and detects unusual activity that may indicate an insider threat or compromised account.

What makes it different:

Instead of detecting one suspicious event, it analyzes the **user's overall behavior pattern**.

Example:

Normal: 9 AM login → normal files → office network

Suspicious: 2 AM login → new device → unusual database → huge file download → external transfer

AI/ML: Isolation Forest, Autoencoders, Random Forest/XGBoost, LSTM.

Novelty Possibility:
Adaptive user profiles + behavioral anomaly detection + explainable alerts + distinguishing compromised account vs insider threat.

Difficulty: High
Potential: Strong cybersecurity + ML research project.

QUICK COMPARISON

Idea	Novelty	Difficulty	AI/ML	Cybersecurity	Potential
Social Engineering Detector	High	Medium-High	★★★★★	★★★★★	🔥🔥🔥
Prompt Injection Firewall	Very High	High	★★★★★	★★★★★	🔥🔥🔥
Deepfake Identity Detector	High	Very High	★★★★★	★★★★★	🔥🔥🔥
Insider Threat Detection	High	High	★★★★★	★★★★★	🔥🔥

CURRENT PREFERENCE

- 1. Prompt Injection Firewall** — Most cutting-edge / research-oriented
- 2. Social Engineering Detector** — Best balance of novelty, feasibility and real-world usefulness
- 3. Deepfake Identity Detector** — Most impressive but hardest
- 4. Insider Threat Detection** — Strong academic cybersecurity project

NEXT STEP

Before selecting one, we should research:

**Existing solutions → Research papers → Datasets → Current limitations → Research gap
→ Our possible novelty → Feasibility**

Then select the idea with the strongest combination of **novelty + feasibility + research value + real-world usefulness**.